IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Digital Object Identifier

When Agreement Fails: Visual Anchoring and Multimodal Pseudo-Label Reliability in Classroom Behavior Recognition

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This work was supported by the Education Department of Hunan Province under Grant HNJG-2022-0608.

ABSTRACT Dual-model agreement is widely used as a pseudo-label reliability heuristic for multimodal large language models (MLLMs), but whether agreement-filtered labels improve downstream fine-tuning in fine-grained classroom behavior recognition remains uncertain. This study presents a reproducible reliability-audit protocol that evaluates agreement filtering under a single-frame protocol on three SCB classroom behavior sub-datasets (BowTurnHead, HandriseReadWrite, TeacherBehavior), linking bounding-box-level MLLM annotation quality measurement, four filtering strategies, repeated-seed CLIP fine-tuning (linear probe and LoRA), retention-ratio diagnostics, selective-routing analysis, and an eleven-model cross-model robustness check spanning seven model families (Qwen2, Qwen2.5, Qwen3.5, Qwen3.6, LLaVA, Gemma3, and Gemma4) across 7B–35B parameter scales.
Three main findings emerge. First, agreement filtering is not a portable denoising rule: on BowTurnHead, agreement-retained linear probing falls to $15.53\% \pm 1.00\%$ while unfiltered pseudo-labels reach $88.02\% \pm 0.12\%$, and retention-ratio controls confirm that this collapse reflects class-composition shift and a label-quality ceiling rather than sample count alone. Second, the operative constraint is visual anchoring—the degree to which a category can be recovered from a single frame without discourse context. Categories with low zero-shot CLIP accuracy (e.g., *guide*, *answer*) systematically fail across all tested MLLMs; newer-generation models (Qwen3.5-27B, Gemma4-31B) substantially improve on *guide* (53% and 83% respectively) while *answer* remains below 23% universally, identifying a task-level single-frame observability boundary. Third, a quality-threshold experiment with Qwen3.5-27B pseudo-labels (annotation accuracy 50.3% vs. Qwen2-VL-7B's 41.2%) locates the critical annotation accuracy range between 41% and 50% at which pseudo-label fine-tuning crosses the zero-shot baseline (55.29% vs. 53.43%). Together, these findings show that downstream utility is gated by per-class visual anchoring and annotation quality thresholds, which provide a task-internal screening signal for deciding when MLLM pseudo-labels are likely to provide useful supervision in classroom behavior recognition.

INDEX TERMS classroom behavior recognition, multimodal large language models, pseudo-label filtering, visual anchoring

I. INTRODUCTION

AUTOMATIC recognition of classroom behaviors—teacher instructional actions, student engagement postures, and off-task micro-movements—is a prerequisite for scalable educational quality assessment and intelligent classroom analytics [1], [2]. Supervised deep learning models have demonstrated strong recognition performance on this task [3], [9], but they require large quantities of labeled images that are labor-intensive to collect: a trained annotator must watch each video clip, identify every person in the frame, draw bounding boxes, and assign the correct activity label, often under ambiguous multi-label conditions. This annotation bottleneck severely limits the scale and diversity of available classroom datasets, and motivates the search for lower-cost labeling alternatives.

Multimodal large language models (MLLMs) such as

Qwen [4] and LLaVA [5] have recently demonstrated impressive visual understanding across a wide range of recognition tasks. Given an image and a list of candidate categories, an MLLM can produce a category prediction without any task-specific training, making them practical candidates for automatic annotation pipelines. Several recent studies have explored LLM-assisted annotation in text-heavy domains [6], [7], and the LATTECLIP framework [8] has shown that MLLM-synthesized text descriptions can drive unsupervised CLIP fine-tuning in general visual domains. However, whether MLLMs can serve as reliable annotators for *fine-grained, domain-specific* behaviors—particularly those requiring interpretation of pedagogical intent rather than surface visual features—has not been systematically studied.

Classroom activity recognition poses distinctive challenges that make this question non-trivial. Some behaviors have clear visual signatures: a teacher writing on a blackboard produces a characteristic image that any vision model should recognise. Others are intent-dependent: the visual difference between a teacher ''answering a student's question'' and a teacher ''giving a lecture'' may be imperceptible from a single frame. This makes it plausible that MLLM annotation quality varies substantially across categories, and that a simple pipeline of ''annotate everything, then fine-tune'' may be undermined by a small number of systematically mis-labelled classes. This paper refers to that category-specific degree of visual identifiability as *visual anchoring*. Formally, visual anchoring is defined as the degree to which a category label can be recovered from a single cropped frame without recourse to discourse context, temporal information, or speaker intent. The concept is used here as a task-internal diagnostic framework, not as an independent causal measure. The operational proxy employed in this study—protocol-matched per-class zero-shot CLIP accuracy—is therefore interpreted as a screening signal within this evaluation setup rather than as a universal quantification of anchoring.

The practical motivation for studying agreement filtering deserves explicit statement. When two independently queried MLLM annotators assign the same label to a crop, the probability that both are correct is generally higher than for a single annotator. This intuition—drawn from multi-annotator agreement in crowdsourcing [32] and consistency-based semi-supervised learning [31]—has made dual-model agreement a widely adopted proxy for pseudo-label reliability. However, this heuristic assumes that annotator errors are independent, which may not hold when both models share architectural biases or when the image itself lacks the visual evidence needed for the task. Understanding when this assumption fails, and what the downstream consequences are, is the central empirical question of this study.

From an engineering perspective, the practical question is not whether MLLMs can produce pseudo-labels at all, but when those labels are reliable enough to improve downstream adaptation and when filtering removes the very supervision the model needs.

Against this background, the study addresses one central question: *when can MLLM-generated pseudo-labels provide a reliable training signal for classroom behavior recognition, and when does agreement filtering make that signal worse rather than better?*

Answering that question is operationalised through four linked analysis modules rather than a chronological phase narrative. First, a protocol-matched zero-shot CLIP baseline establishes the training-free reference point under the same bounding-box crop setting. Second, MLLM annotation and validation measure category-level annotation reliability and test whether the observed failure modes depend on the original annotator pair. Third, CLIP adaptation under label-source choices compares unfiltered pseudo-labels, agreement-filtered pseudo-labels, and ground-truth supervision across linear probing and LoRA. Fourth, mechanism and strategy audits use retention controls, anchor-aware selective routing, auxiliary filtering and self-training baselines, and cross-model consistency checks to distinguish between data-volume loss, sampling bias, category semantics, training-strategy effects, and annotator dependence.

The main pseudo-label annotation pipeline uses two MLLMs (Qwen2-VL-7B-Instruct and LLaVA-1.5-7B) to annotate cropped bounding-box regions from three SCB sub-datasets covering 13 behavior categories across more than 87,000 bounding boxes. An eleven-model robustness analysis (expanded from five models in the original submission) is treated as a boundary condition check, not as the paper's main claim. The core answer comes from the downstream fine-tuning matrix, while the additional diagnostics explain why that answer changes across datasets and categories.

The contributions are fourfold, with the audit protocol as the main technical object rather than any single routing heuristic.

(1) **A reproducible reliability-audit protocol for MLLM pseudo-labeling.** The protocol connects bbox-level annotation accuracy, repeated-seed downstream CLIP adaptation, retention-ratio controls, distribution-shift checks, auxiliary training-strategy baselines, anchor-proxy analysis, and cross-model robustness.
(2) **Pseudo-label usefulness is category-conditional rather than global.** Visual anchoring separates classes that MLLMs can label reliably from classes that remain structurally ambiguous under single-frame crops.
(3) **Agreement filtering is not a safe denoising default.** The diagnostic evidence shows that agreement-filtered subsets can be affected by data-volume loss, class-composition shift, and shared label error, even when retained-subset annotation accuracy appears to improve.
(4) **The boundary conditions are separated from a single annotator choice or optimisation path.** An oracle selective-routing diagnostic, targeted LoRA replications, auxiliary teacher–student and confidence-filtering baselines, and a eleven-model robustness check bound the claim and point to label quality

and category semantics as the main constraints in the present setting.

II. RELATED WORK

A. CLASSROOM ACTIVITY RECOGNITION

Automated analysis of classroom teaching and learning behaviors has been approached through video [10], audio [11], and multimodal sensing [12]. CNN-based methods achieved early success on student engagement detection [3], and transformer architectures have more recently enabled richer spatiotemporal modelling [9]. A persistent challenge across all these works is the scarcity of labeled data: classroom recordings raise privacy concerns, annotation requires domain expertise, and activity categories are often defined by pedagogical intent that is not directly visible in individual frames. The analysis addresses this bottleneck by evaluating whether MLLMs can substitute for human annotators, thereby reducing labeling cost.

B. VISION–LANGUAGE MODELS AND ZERO-SHOT TRANSFER

CLIP [13] established contrastive image–text pretraining as a powerful paradigm for zero-shot visual recognition, enabling classification by matching image embeddings to natural language category descriptions. Subsequent work has extended this framework through improved training data [14], [15], alternative loss functions [16], masked image modeling [17], and data filtering networks [18]. Domain-specific applications include medical imaging [19], remote sensing [20], and action recognition [21]. Zero-shot CLIP performance is known to be sensitive to prompt design [13], and methods such as CoOp [22] and CoCoOp [23] learn soft prompt embeddings to adapt CLIP to target domains with minimal supervision. In the classroom domain, however, zero-shot CLIP performance across multiple scene types and model variants remains understudied.

C. LLM- AND MLLM-ASSISTED DATA ANNOTATION

Using language models to generate or refine training labels has attracted growing interest. Early work showed that GPT-3/4-class models can produce competitive NLP annotations for tasks such as stance classification and named entity recognition [6], [7]. In the multimodal setting, Zhang et al. [24] benchmark zero-shot MLLM annotation for facial expression images, finding large variance across model families and expression categories—a pattern that parallels the results reported here. However, that study stops at annotation benchmarking; it does not test downstream CLIP adaptation, compare filtering protocols, or analyse why particular categories fail. Most relevant here is LATTECLIP [8], which uses MLLM-generated rich textual descriptions to fine-tune CLIP without any human labels in a general visual setting; the authors note that annotation quality is especially low for classes that are "not visually discriminative," foreshadowing the visual anchoring finding. Recent open-weight MLLM progress in 2024–2025 has also changed the practical annotation landscape. Model families such as LLaVA-NeXT, Qwen2.5-VL, and InternVL2 report stronger general visual reasoning and instruction following capabilities [25]–[27], while broader multimodal benchmarks such as MMMU still show that domain-specific reasoning remains uneven and task dependent [28]. This is directly relevant to pseudo-label pipelines because better average MLLM capability does not automatically imply reliable category coverage under constrained prompts and single-frame evidence. The work differs from LATTECLIP in three ways: it operates in a fine-grained domain-specific setting, uses bounding-box-level rather than image-level annotations, and explicitly characterises the relationship between per-category annotation accuracy and downstream fine-tuning performance. More broadly, the paper combines three threads that are usually treated separately: MLLM-based automatic annotation, pseudo-label filtering, and classroom behavior recognition. To the best of current knowledge, no prior work has systematically evaluated MLLM annotation reliability for classroom activity recognition or provided per-category diagnostic evidence of annotation failure modes.

D. PSEUDO-LABEL FILTERING STRATEGIES

Filtering noisy pseudo-labels is a standard component of semi-supervised and weakly supervised learning pipelines [29], [30]. Common strategies include confidence thresholding, consistency regularization across augmentations, and ensemble agreement [31]. In the classic semi-supervised setting, however, consistency filtering usually compares predictions from the same model under perturbation or augmentation, whereas this work examines agreement between two different MLLMs on the same crop. That distinction matters because cross-model agreement can remove not only noisy samples but also large, systematically informative regions of the training distribution when the two annotators fail in different ways. The agreement-based filter used here—retaining samples where two independently queried MLLMs assign the same label—is conceptually related to multi-annotator agreement in crowdsourcing [32], where consensus labels are generally higher quality but fewer in number. Another major family of pseudo-label filtering methods uses calibrated confidence or uncertainty estimates instead of agreement alone. That comparison is relevant here, but the executed annotation protocol uses index-only prompting and does not expose a calibrated scalar confidence signal; for that reason, the experiments test only the screening cues actually produced by the annotation pipeline, namely valid-output screening and cross-model agreement. A key finding is that, in the observed retention regime (20–67% on the training splits), the data-volume cost and sampling bias of agreement filtering can overwhelm any quality benefit, leading to substantial over-fitting or negligible downstream gains. In these experiments, this pattern directly contradicts the common intuition that cross-model agreement automatically yields better training data. This complements findings in the NLP annotation literature showing that LLM label quality does not monotonically improve with filtering stringency when the underlying model systematically errs on certain input types [7]. Related 2024 reliability-oriented evaluations in multimodal settings similarly suggest that robust deployment requires explicit uncertainty or error-structure checks before assuming that stricter filtering improves supervision quality [24], [28]. Taken together, the gap in prior work is not the absence of MLLM annotation studies, pseudo-label filtering methods, or classroom recognition benchmarks in isolation, but the lack of a single study that tests these components jointly and explains when and why the combined pipeline fails in a classroom-behavior setting.

TABLE 1. Overview of the three SCB sub-datasets used in this study, including per-class bounding-box counts. Class imbalance interacts with agreement filtering and anchor-aware routing (Sec. V).

Sub-dataset	Class	Train BB	Val BB	Train/Val images
BowTurnHead (2-class)	bow-head	4,422	540	1,905 / 505
	turn-head	7,943	3,213	
HandriseReadWrite (3-class)	hand-raise	10,538	2,915	5,193 / 1,671
	read	17,539	6,539	
	write	6,447	3,394	
TeacherBehavior (8-class)	guide	1,155	449	8,984 / 3,240
	answer	2,574	853	
	on-stage interaction	528	149	
	blackboard-writing	821	277	
	teacher	8,490	3,228	
	stand	13,932	4,967	
	screen	5,025	1,959	
	blackboard	7,847	3,445	

III. DATASETS AND EXPERIMENTAL SETUP

A. PIPELINE OVERVIEW

The experimental pipeline consists of three stages, each producing fixed artifacts consumed by downstream analyses:

(1) **Annotation stage:** YOLO-detected bounding-box crops are annotated by two MLLMs (Qwen2-VL-7B-Instruct and LLaVA-1.5-7B) under an index-only zero-shot prompt. Each crop receives a predicted class index from each model. Validation-set annotations are used only for quality measurement; training-set annotations are exported as JSONL for downstream fine-tuning.

(2) **Filtering stage:** Four filtering strategies are compared: None (all Qwen predictions), Valid-Output-Qwen (Qwen outputs a valid index), Valid-Output-LLaVA (LLaVA outputs a valid index), and Agreement (Qwen and LLaVA assign the same index). Only the None and Agreement strategies produce training pseudo-label JSONL files for the downstream stage.

(3) **Fine-tuning stage:** CLIP ViT-L/14 is fine-tuned under three label sources (None pseudo-labels, Agreement pseudo-labels, GT) and two training modes (linear probe, LoRA) across three SCB sub-datasets, forming an 18-condition main matrix plus three zero-shot baselines. Four diagnostic modules—selective routing, retention-ratio curves, strategy audits (confidence filtering, teacher–student self-training, cross-model consistency), and an eleven-model cross-model robustness check—read frozen artifacts from earlier stages to isolate specific mechanisms.

B. DATASETS

All experiments use three sub-datasets from the publicly available SCB (Smart Classroom Behavior) benchmark [33], which provides bounding-box annotations for classroom activity images captured in real Chinese K–12 classrooms. The three sub-datasets cover complementary scene types and difficulty levels, summarised in Table 1.

Class balance is uneven in all three sub-datasets. BowTurnHead is skewed toward *turn-head*; HandriseReadWrite is dominated by *read*; and TeacherBehavior is dominated by *stand*, *teacher*, and *blackboard*, while *on-stage interaction* and *blackboard-writing* are rare. These imbalances provide relevant context for interpreting agreement retention bias and selective-routing behavior.

BowTurnHead contains images of students exhibiting two micro-action classes: *bow-head* and *turn-head*. The *bow-head* class typically corresponds to a downward head posture associated with distraction or phone use. The binary nature of the task and the visually distinct postures make this the easiest sub-dataset.

HandriseReadWrite contains three student behavior classes: *hand-raise*, *read*, and *write*. These classes require observing upper-body posture and hand position, introducing moderate visual ambiguity between reading and writing.

TeacherBehavior contains eight teacher activity classes: guide, answer, on-stage interaction, blackboard-writing, teacher (general presence), stand, screen, and blackboard. The large class count, overlapping visual appearances, and the prevalence of intent-dependent categories (e.g., "answer" versus "guide" are visually nearly identical) make this the most challenging sub-dataset.

Throughout the manuscript, class names follow the index order used in the annotation prompt and evaluation protocol: BowTurnHead = {0: *bow-head*, 1: *turn-head*}; HandriseReadWrite = {0: *hand-raise*, 1: *read*, 2: *write*}; TeacherBehavior = {0: *guide*, 1: *answer*, 2: *on-stage interaction*, 3: *blackboard-writing*, 4: *teacher*, 5: *stand*, 6: *screen*, 7: *blackboard*}.

All annotations are provided in YOLO format (class index, normalised centre coordinates, width, height). Each bounding box defines a single activity instance; images may contain multiple bounding boxes corresponding to different individuals or simultaneous activities.

C. ETHICS AND DATA GOVERNANCE

This study is a secondary analysis of a previously released public benchmark of classroom images and bounding-box annotations. The authors did not recruit participants, collect new data, intervene in classroom activities, or access metadata beyond the released image and label files. The analysis uses only the released benchmark files, reports aggregate results, and does not reproduce identifiable classroom images. Under this secondary-analysis design, additional institutional review and informed consent were not required for the present study.

Ethical oversight for the original classroom data collection remains with the SCB dataset creators and the institutions that approved its release. The present work relies on those original approvals and follows the authors' institutional policy for secondary analysis of public child image datasets: no re-identification attempts, no release of new identifiable child images, and aggregate-only reporting. All qualitative examples in the manuscript are privacy-preserving pixelated crops generated for illustration, and the study does not redistribute raw classroom images outside the public dataset release.

D. MLLM ANNOTATION PIPELINE

1) Bounding Box Cropping

For each annotated bounding box, the corresponding image region is cropped with a 5% margin on each side to provide contextual information around the target. Crops smaller than 32×32 pixels after margin expansion are discarded. This yields 87,261 training crops and 31,928 validation crops across the three sub-datasets.

2) Models

In the main pseudo-label annotation pipeline, two locally deployable MLLMs serve as annotators: **Qwen2-VL-7B-Instruct** [4] (hereafter Qwen) and **LLaVA-1.5-7B** [5] (hereafter LLaVA), both running on dedicated GPUs to enable parallel annotation. The study selects locally deployable 7B-scale models for three reasons: reproducibility without API dependency, cost-free inference at scale, and comparability between the two models under identical hardware conditions.

3) Prompt Design

Both models receive a uniform zero-shot prompt listing the category names with their indices, followed by the instruction to respond with the index only:

> *You are a classroom behavior recognition expert. Given one cropped classroom image, choose the single best matching class. Reply with only the class index number. No explanation. Classes: [0: class_0, 1: class_1, …]. Answer with index only:*

The same prompt serves all three sub-datasets; only the class list changes. No dataset-specific synonyms, textual definitions, or auxiliary class descriptions are added beyond the class names shown in the prompt. This index-only format standardises the output space across models and reduces parsing ambiguity, but it does not measure uncertainty and may force low-confidence decisions into a single discrete label. For all executed annotation runs, decoding is deterministic (`do_sample=False`), and the response length is capped at `max_new_tokens=8`.

Table 2 summarises the label-index order used in the executed runs. The same dataset-specific class list is passed to both MLLMs from a shared dataset configuration, and downstream evaluation reuses the same order for label decoding. Valid model outputs are single in-range integers such as "0", "1", or "4"; any non-integer or out-of-range string is recorded as invalid instead of coerced to a class label.

TABLE 2. Label-index order audit for the index-only prompt. The same ordering is used for both MLLM annotation and downstream label decoding, providing a direct consistency check against index-mapping errors.

Dataset	Index order used in prompt and evaluation
TeacherBehavior	0: guide; 1: answer; 2: on-stage interaction; 3: blackboard-writing; 4: teacher; 5: stand; 6: screen; 7: blackboard
HandriseReadWrite	0: hand-raise; 1: read; 2: write
BowTurnHead	0: bow-head; 1: turn-head

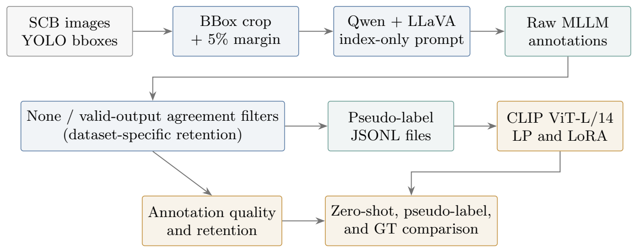

FIGURE 1. Experimental pipeline for MLLM-assisted classroom behavior annotation. Bounding-box crops are annotated by two 7B-scale MLLMs, converted into pseudo-label sets through filtering strategies, and used to fine-tune CLIP with linear probing or LoRA. The same pipeline also yields annotation-quality and retention diagnostics.

Fig 1 summarizes the full experimental pipeline, from bbox cropping and MLLM annotation to pseudo-label filtering, CLIP fine-tuning, and diagnostic interpretation.

4) Filtering Strategies

For implementation audit purposes, the pipeline records four filtering variants, but only two of them represent substantively different selection rules under the executed protocol:

(1) **None (unfiltered):** All crops for which Qwen produces a valid index are retained. The pipeline uses the Qwen prediction as the pseudo-label.

(2) **Valid-Output-Qwen:** Only crops where Qwen outputs a valid in-range index are retained (identical to None in practice, as Qwen has a near-100% valid-output rate).

(3) **Valid-Output-LLaVA:** Only crops where LLaVA outputs a valid index are retained.

(4) **Agreement:** Only crops where Qwen and LLaVA assign the same index are retained; crops with disagreement are excluded, and the shared index is used as the pseudo-label.

Because the annotation prompt enforces an index-only response and does not expose calibrated confidence scores, confidence thresholding is not available as a distinct scalar filter in the executed runs. In practice, Valid-Output-Qwen collapses to None and Valid-Output-LLaVA differs only through valid-output screening, so the downstream analysis focuses on None and Agreement as the two informative representative strategies. The goal of this comparison is not to optimise a confidence threshold, but to test whether the readily available screening signals produced by the executed annotation protocol—valid-output screening and cross-model agreement—improve over unfiltered pseudo-labels under the same prompt and crop setting.

E. CLIP FINE-TUNING

1) Backbone

CLIP ViT-L/14 with OpenAI pretrained weights [13] serves as the visual backbone throughout, matching the model family used in prior zero-shot evaluation on these datasets [33]. The implementation uses `open_clip_torch` version 2.32.0. The image encoder produces 768-dimensional embeddings.

2) Fine-Tuning Methods

Two parameter-efficient fine-tuning strategies are compared:

Linear Probe (LP): The CLIP visual encoder is fully frozen. A single linear classification head ($768 \rightarrow C$, where C is the number of classes) is trained on the pseudo-labeled crops using cross-entropy loss.

Low-Rank Adaptation (LoRA): LoRA layers [34] with rank $r = 8$ and scaling factor $\alpha = 16$ are injected into the two MLP layers (`mlp.c_fc` and `mlp.c_proj`) of all transformer blocks in the visual encoder. LoRA layers are injected only into the MLP projections, as replacing the `MultiheadAttention` output projection in the `open_clip` implementation interfered with direct access to the underlying `out_proj.weight` and `out_proj.bias` attributes. This introduces 1.97M LoRA parameters while keeping the remaining ~303M CLIP parameters frozen. The linear classification head is trained jointly with the LoRA parameters.

3) Training Details

All models are trained for 20 epochs using AdamW optimiser (lr $= 10^{-4}$, weight decay $= 0.01$) with cosine annealing learning rate schedule. The reported main-matrix and targeted-replication runs use batch size 128, distributed across two NVIDIA A100-SXM4-40GB GPUs via `DataParallel`. No gradient accumulation is used, so the effective optimisation batch size equals the per-step global batch size specified in each run. Best model selection is based on validation accuracy. For the main downstream matrix, LP conditions are averaged over repeated seeds (all datasets: five seeds, 43–47). A targeted LoRA replication over five seeds (43–47) is reported for BowTurnHead and TeacherBehavior under None and GT; the remaining LoRA values stay as protocol-matched single-run evaluations (seed 42). Retention-ratio and selective-routing diagnostics are reported as fixed-seed runs (seed 42). An overview of seed usage by module is provided in Table 3. Pseudo-label generation and CLIP fine-tuning remain split-isolated: validation crops are used only for annotation evaluation and model selection, whereas pseudo-label files consumed by CLIP fine-tuning are generated exclusively from the training split.

4) Experimental Conditions

For each of the three sub-datasets and two fine-tuning methods, the study evaluates three pseudo-label sources:

- **None:** Unfiltered MLLM pseudo-labels (Qwen predictions).

TABLE 3. Seed policy by analysis module. Main-matrix LP uses repeated-seed evaluation; targeted LoRA replication is reported for BowTurnHead and TeacherBehavior under None and GT; auxiliary diagnostics and the remaining LoRA reference cells remain protocol-matched single-run/fixed-seed analyses.

Module	Protocol scope	Seed policy
Main downstream matrix (LP)	3 datasets × 3 label sources	5 seeds (43–47)
Targeted LoRA replication	BowTurnHead + TeacherBehavior; None/GT	5 seeds (43–47)
Main downstream matrix (remaining LoRA)	HandriseReadWrite + agreement conditions	single run (42)
Retention-ratio curves	3 datasets, LP diagnostics	fixed seed (42)
Selective routing (Scheme B)	TeacherBehavior oracle diagnostic	fixed seed (42)
Auxiliary filtering/self-training baselines	3 datasets, LP diagnostics	5 seeds (43–47)
Cross-model annotation and consistency check	5 MLLMs annotation quality	deterministic eval protocol

- **Agreement:** Agreement-filtered pseudo-labels.
- **GT:** Ground-truth human annotations (supervised reference).

This yields $3 \times 2 \times 3 = 18$ experimental conditions in total. The zero-shot CLIP baseline (no fine-tuning, prompt-engineered text embeddings) provides the primary practical reference. Prompt-adaptation methods such as CoOp and CoCoOp are not included as direct baselines here because they require target-domain optimisation; this comparison isolates the value of pseudo-label supervision against a training-free reference and a ground-truth upper bound under the same crop protocol. The main-result zero-shot values (Table 7) are obtained under the same bbox-crop validation protocol using the canonical zero-shot evaluation setting. Per-class prompt-ensemble outputs from the same evaluation pipeline are used separately in the anchor-proxy analysis and the selective-routing experiment.

F. INTEGRATED DIAGNOSTIC ANALYSES

Beyond the main annotation-quality and downstream fine-tuning comparisons, the study design includes linked diagnostic analyses. Each addresses a different decision point in the pseudo-labeling workflow: whether categories are visually grounded enough for automatic annotation, whether agreement filtering fails because of retention loss or sampling bias, whether category-aware routing can serve as an upper-bound diagnostic on the hardest taxonomy, whether simple confidence-filtering or self-training alternatives change the downstream interpretation, and whether the main annotation patterns remain stable under model substitution. Together, these analyses are intended to explain the main findings, not to replace repeated-seed downstream validation.

1) Selective Annotation (Scheme B)

To validate the diagnostic utility of visual anchoring as an annotation routing criterion, the study runs a selective annotation experiment on TeacherBehavior. The four high-anchor categories—*blackboard-writing*, *teacher*, *screen*, and *blackboard*—are trained using unfiltered pseudo-labels (None strategy) with a linear probe classification head restricted to those four classes. This selective routing is evaluated only as an oracle upper-bound diagnostic, not as a directly deployable pipeline. The four low-anchor categories—

guide, *answer*, *on-stage interaction*, and *stand*—are not included in fine-tuning; at inference time, these categories are handled by the frozen zero-shot CLIP classifier built with a prompt ensemble. During offline evaluation, the final prediction for each validation crop selects the fine-tuned head for high-anchor inputs and the zero-shot classifier for low-anchor inputs using the ground-truth category only as an oracle routing variable. The resulting overall 8-class accuracy should therefore be interpreted strictly as an upper-bound diagnostic for category-aware routing, not as a directly deployable inference pipeline. A deployable variant would need a separate routing predictor derived from observable cues or proxy scores instead of ground-truth labels.

Pilot implementation tests with LoRA showed rapid overfitting and severe degradation of the low-anchor zero-shot branch, because LoRA modifies the shared visual encoder used by both branches. For this reason, Scheme B reporting focuses on linear probing.

2) Retention-Ratio Curve

To characterise the relationship between training data volume and downstream performance more precisely than the two-endpoint comparison afforded by the None versus Agreement conditions, the study evaluates intermediate retention ratios of 25%, 50%, and 75% on BowTurnHead and HandriseReadWrite. For each ratio, a random subset of the corresponding size is drawn from the unfiltered pseudo-label training set (fixed seed 42), and a linear probe is trained for 20 epochs under identical hyper-parameters. The resulting five-point curves (25%, 50%, 75%, 100% from step3, and the Agreement endpoint) provide a finer-grained view than a simple two-endpoint comparison. These retention-ratio diagnostics intentionally remain fixed-seed comparisons so that data-volume effects can be read against a single controlled subset draw; the main LP results in Table 7 instead report repeated-seed averages (all datasets: five seeds, 43–47).

3) Auxiliary Filtering and Teacher–Student Baselines

Two auxiliary training-strategy baselines test whether simple alternatives to the main pseudo-label protocol change the downstream interpretation. The confidence-filtering baseline starts from the unfiltered Qwen pseudo-label training set. Each training crop is encoded with CLIP ViT-L/14, the same prompt-ensemble zero-shot classifier is constructed, and the CLIP probability assigned to the Qwen pseudo-label is used as a confidence score. Linear probes are then trained on the retained samples for thresholds $\{0.3, 0.5, 0.6, 0.7, 0.8, 0.9\}$ across seeds 43–47. Because the MLLM prompt uses index-only outputs, this score is a CLIP-assisted confidence proxy, not calibrated MLLM logit confidence.

The teacher–student baseline evaluates a label-scarce self-training alternative under the same CLIP feature protocol. For each seed, a linear-probe teacher is trained on a stratified 10% labeled subset of the training split. The teacher then assigns pseudo-labels to the remaining training crops, and a student linear probe is trained on those teacher labels together with the labeled training subset. Validation labels are not used as training labels; they are used only for model selection and evaluation under the same validation protocol as the main LP experiments. This baseline is a label-scarce reference for semi-supervised training under limited human annotation, not a label-free alternative to the MLLM pseudo-labeling pipeline.

4) Cross-Model Robustness Boundary Check

To test whether the main annotation findings depend strongly on the original two-model main pipeline (Qwen2-VL-7B and LLaVA-1.5-7B), the revised study includes an expanded cross-model robustness check with eleven open-weight MLLMs spanning seven model families (Qwen2, Qwen2.5, Qwen3.5, Qwen3.6, LLaVA, Gemma3, and Gemma4), three architectural types (dense, MoE, and 4-bit quantized), and parameter scales from 7B to 35B: Qwen2-VL-7B-Instruct, LLaVA-1.5-7B, Qwen2.5-VL-7B-Instruct, Qwen2.5-VL-32B-Instruct, Qwen3.5-27B, Qwen3.5-35B-A3B (MoE), Qwen3.6-27B, Qwen3.6-35B-A3B (MoE, FP8), Gemma-3-27B-IT (4-bit), Gemma4-26B-A4B-it (MoE), and Gemma4-31B-it. All runs use the same bbox-crop protocol, index-only prompt, and label decoding as the main annotation pipeline. Validation annotations are collected for all three sub-datasets; TeacherBehavior is additionally processed on the training split to check whether the same category-level failure pattern persists outside the validation set. The cross-model analysis focuses on per-model pseudo-label accuracy and on pairwise agreement-subset accuracy among model pairs, and the expanded model set allows testing whether the observed failure patterns generalise across model generations, architectures, and scales. This robustness check strengthens external validity with respect to annotator choice, but it does not by itself resolve the within-protocol variance question created by the remaining single-run downstream settings.

5) Operational Proxy for Visual Anchoring

To make visual anchoring operational here, the analysis defines a category-level proxy score using protocol-matched per-class zero-shot CLIP accuracy:

$$\mathrm{AnchorProxy}(y) = \mathrm{Acc}_y^{\mathrm{ZS}}.$$

Here, $\mathrm{Acc}_y^{\mathrm{ZS}}$ denotes the per-class accuracy of zero-shot CLIP (prompt-ensemble setting) under the same bbox-crop validation protocol used for all downstream experiments. Higher values indicate that category y is more visually anchored, whereas lower values indicate stronger dependence on intent or context. This proxy supports diagnostic interpretation and is not claimed as a causal estimator or as an independent ground-truth measure of visual anchoring. Because the proxy is derived from another classifier's observed performance, it should be interpreted as an operational screening cue within this evaluation setup, not as an independent explanatory measurement. As a convergent diagnostic, the study also reports a class-level cross-model consistency (CMC) score computed

TABLE 4. MLLM pseudo-label annotation accuracy and retention rate on the validation split. "Pseudo-label acc." = percentage of pseudo-labels matching the ground-truth label for retained bbox crops. Under the None strategy, "Retained" is the number of crops with a valid Qwen index; Qwen produces valid outputs for more than 99.9% of validation crops. These retention rates are computed on the validation split only; training-set retention rates used later in the retention-ratio analysis are reported separately in the text and figure.

Sub-dataset	None (Qwen)		Agreement (Qwen+LLaVA)		
	Pseudo-label acc. (%)	Retained	Pseudo-label acc. (%)	Retained	Ret. rate (%)
BowTurnHead	82.55	3,753	41.76	613	16.3
HandriseReadWrite	70.60	12,843	73.97	7,122	55.4
TeacherBehavior	41.21	15,327	47.57	10,167	66.3

from the eleven MLLM annotators. For each ground-truth category, CMC is the mean pairwise agreement among model predictions on validation crops from that category. CMC is independent of the CLIP zero-shot classifier used to construct $\mathrm{AnchorProxy}$, but it is still treated as a descriptive consistency signal, not as a causal measure of anchoring. In addition, because Scheme B routes low-anchor classes through the same zero-shot CLIP branch used to construct $\mathrm{AnchorProxy}(y)$, these two signals are operationally coupled here and should not be interpreted as independent confirmatory evidence.

IV. RESULTS

Results are organised around the paper's central question, not around the repository's execution order. The section first establishes where MLLM pseudo-labels are reliable enough to serve as plausible training signal, using annotation-quality and per-category evidence. It then tests whether those pseudo-labels actually help downstream CLIP adaptation relative to zero-shot inference and the ground-truth reference. Finally, diagnostic and robustness analyses delimit why agreement filtering fails and how strongly the conclusions depend on annotator choice.

A. MLLM ANNOTATION QUALITY

Table 4 reports pseudo-label accuracy and retention rates for the None and Agreement strategies on the validation split of each sub-dataset. Each retained bounding-box crop is treated as one evaluation instance, so both accuracy and retention are defined at bbox level, not at whole-image level.

Three observations follow. First, annotation accuracy decreases monotonically with task complexity: from 82.6% for the binary BowTurnHead to 41.2% for the eight-class TeacherBehavior. Second, Agreement filtering is not a uniformly reliable default: it improves retained-subset accuracy on HandriseReadWrite and TeacherBehavior, but substantially reduces both retention and accuracy on BowTurnHead. Only 16.3% of BowTurnHead validation crops survive the agreement filter, and their accuracy falls from 82.55% to 41.76%. Third, even after filtering, annotation accuracy remains well below the ground-truth reference for HandriseReadWrite and TeacherBehavior. To avoid ambiguity, the manuscript distinguishes these validation-set retention rates (16.3%, 55.4%, and 66.3%) from the training-set retention rates used in downstream fine-tuning (20.0%, 55.9%, and 66.5%).

TABLE 5. BowTurnHead validation-set confusion counts for Qwen and LLaVA. Rows denote ground-truth labels; columns denote model predictions.

GT class	Qwen→bow	Qwen→turn	LLaVA→bow	LLaVA→turn
bow-head	237	303	534	6
turn-head	352	2,861	3,193	20

TABLE 6. Per-category MLLM annotation accuracy on TeacherBehavior (Agreement strategy, validation split). Accuracy is reported to two decimal places to match the other quantitative tables.

Category	Accuracy (%)	n
blackboard-writing	98.61	216
teacher	92.01	2,729
screen	69.41	1,154
blackboard	57.42	2,238
answer	0.00	415
guide	0.00	327
on-stage interaction	0.00	30
stand	0.85	3,058

Pairwise cross-model agreement follows the same pattern. Because both models produce valid outputs for almost every validation crop, the agreement rate is 16.33% on BowTurnHead, 55.46% on HandriseReadWrite, and 66.33% on TeacherBehavior, leaving corresponding disagreement rates of 83.67%, 44.54%, and 33.67%. BowTurnHead is the clearest failure case: the two models disagree on 3,140 of 3,753 validation crops, so the agreement filter keeps only a small and highly selective subset, not a uniformly cleaner one.

Table 5 clarifies why agreement filtering collapses on BowTurnHead. Qwen predicts *turn-head* on most *turn-head* crops, whereas LLaVA collapses almost the entire *turn-head* class into *bow-head*. The dominant disagreement pattern is ground-truth *turn-head* with Qwen = *turn-head* and LLaVA = *bow-head* (2,842 crops), followed by ground-truth *bow-head* with Qwen = *turn-head* and LLaVA = *bow-head* (297 crops). Within the 613 agreed crops, the models jointly predict only 19 *turn-head* instances correctly but jointly mislabel 351 *turn-head* instances as *bow-head*, so agreement selects shared bias, not shared correctness.

1) Per-Category Analysis of TeacherBehavior

Table 6 presents per-category annotation accuracy under the Agreement strategy for TeacherBehavior, revealing a clear bimodal distribution.

The upper group comprises categories with *strong visual anchors*: each label corresponds to a distinctive, self-contained visual element (chalk marks on a blackboard; a projection screen; the blackboard surface itself). The lower group comprises categories whose labels are determined by *behavioural intent* more than visual appearance: "answer" and "guide" are visually indistinguishable from a general lec-

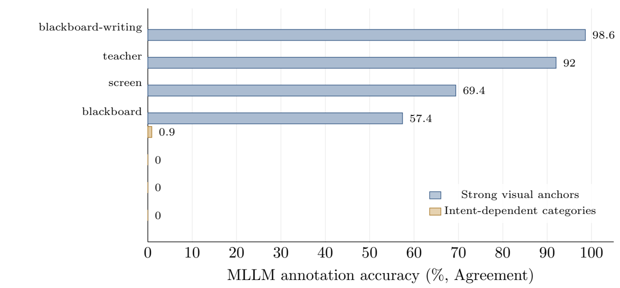

FIGURE 2. **Per-category MLLM annotation reliability on TeacherBehavior under the Agreement strategy. Categories with explicit visual anchors are annotated reliably, whereas categories requiring intent inference drop to near-zero accuracy in the reported setting.**

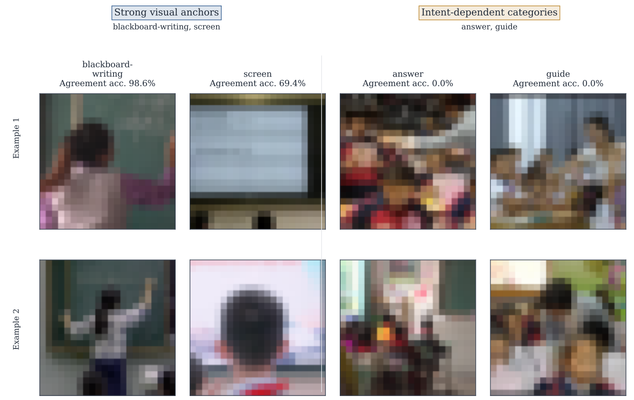

FIGURE 3. **Privacy-preserving pixelated crops from the TeacherBehavior validation agreement subset. High-anchor classes are associated with localized visual cues, whereas low-anchor intent-defined classes remain visually close to generic classroom interaction. The agreement accuracies shown above each category match Table 6.**

ture posture; "stand" is present in nearly every frame regardless of the actual activity. The exact 0.00% values for *answer*, *guide*, and *on-stage interaction* therefore indicate a systematic failure on intent-defined classes, not a rounding artifact. An index-order error would be expected to induce a stable permutation pattern across all classes and analysis modules, but the observed errors are class-specific and asymmetric instead. The error flows are also highly structured. Across all 853 validation instances of *answer*, Qwen maps 831 to *teacher*, while LLaVA splits the same class mainly between *teacher* (420) and *guide* (363); the agreement subset contains 415 crops and none is correct. Across 449 *guide* instances, Qwen predicts *teacher* 444 times and LLaVA predicts *teacher* 330 times, again yielding 0.00% agreement accuracy. For *on-stage interaction*, Qwen predicts *teacher* on 130 of 149 crops, whereas LLaVA predicts *on-stage interaction* on 84 of 149 crops but rarely agrees with Qwen; only 30 crops enter the agreement subset and all 30 are wrong. These patterns support a semantic-confusion explanation, not an indexing error. The BowTurnHead confusion counts in Table 5 and the class-specific error flows reported here provide the direct audit evidence for that interpretation. This category-level distinction is referred to throughout the analysis as *visual anchoring*.

Fig 2 visualizes this separation. Categories with strong, object-like visual anchors occupy the high-accuracy group, whereas intent-dependent teacher behaviors form a near-zero-accuracy failure group.

Privacy-preserving representative crops in Fig 3 make the same contrast concrete. Classes such as *blackboard-writing* and *screen* are tied to localized objects or surfaces that remain visually specific within one crop, whereas *answer* and *guide* reduce to generic teaching or teacher-student interaction postures despite carrying different semantic labels.

B. FINE-TUNING RESULTS

Table 7 presents the complete $3 \times 3 \times 2$ result matrix together with the zero-shot CLIP baseline, while marking which LoRA cells are targeted replications and which remain single-run protocol-matched references.

Table 8 reports two auxiliary training-strategy baselines under the same CLIP ViT-L/14 feature protocol. Confidence

TABLE 7. **Validation accuracy (%) of CLIP ViT-L/14 under all experimental conditions. LP values report mean ± SD over repeated seeds (all datasets: five seeds, 43–47); LoRA values report mean ± SD for the targeted five-seed replication conditions (BowTurnHead and TeacherBehavior under None and GT; seeds 43–47) and seed-42 protocol-matched references elsewhere. Bold marks the best repeated-seed LP result per dataset; single-run LoRA references marked with † are not included in best-result marking. ZS = zero-shot CLIP baseline (no fine-tuning). LP = linear probe. LoRA = low-rank adaptation. These repeated-seed LoRA cells correspond only to those four replicated conditions; see Finding 5 for the associated curve analysis. The main conclusions about agreement filtering are drawn primarily from repeated-seed LP results and annotation/retention diagnostics; single-run LoRA cells are reported as protocol-matched references.**

Dataset	ZS	None		Agreement		GT	
		LP	LoRA	LP	LoRA	LP	LoRA
BowTurnHead	42.37	88.02±0.12	88.65±0.52	15.53±1.00	19.69†	**93.35±0.08**	97.91±0.10
HandriseReadWrite	56.88	75.64±0.13	76.69†	52.88±0.08	59.25†	**79.43±0.20**	87.06†
TeacherBehavior	37.13	42.38±0.07	43.22±0.46	43.01±0.05	45.18†	**69.56±0.05**	74.71±0.11

filtering approximately reproduces unfiltered pseudo-label LP performance on BowTurnHead and HandriseReadWrite: 87.99% versus 88.02%, and 75.56% versus 75.64%, respectively. However, it produces no usable TeacherBehavior training subset under the evaluated threshold grid, indicating that CLIP-assigned confidence is uniformly low for the Qwen pseudo-labels on this taxonomy; in practice, even the lowest threshold in the grid retained no TeacherBehavior training crops. The teacher–student baseline has a different task-dependent profile. It slightly improves BowTurnHead, underperforms direct pseudo-label supervision on HandriseReadWrite, and raises TeacherBehavior from 42.38% under direct pseudo-label LP to 65.06%, approaching the 69.56% ground-truth LP reference. Because this teacher–student setting uses a stratified 10% labeled subset of the training split, it should be read as a label-scarce semi-supervised reference, not as a label-free pseudo-labeling baseline. Taken together, these baselines reinforce the paper's conditional claim without identifying a uniformly dominant training rule.

Fig 4 visualizes the same result matrix while keeping evidence levels explicit. The figure highlights three trends

TABLE 8. Auxiliary filtering and teacher–student baselines under the same CLIP ViT-L/14 feature protocol. Values report validation accuracy (%) over five seeds (43–47). Confidence filtering reports the best nonempty threshold from the pre-specified grid $\{0.3, 0.5, 0.6, 0.7, 0.8, 0.9\}$; the value in parentheses gives the selected threshold and mean training-set retention.

Baseline	BowTurnHead	HandriseReadWrite	TeacherBehavior
Confidence filtering[‡]	87.99±0.22 ($\tau = 0.3$; 100%)	75.56±0.24 ($\tau = 0.3$; 100%)	–
Teacher–student	89.17±1.06	68.08±0.23	65.06±0.36

[‡]No TeacherBehavior confidence-filtering result is reported because all evaluated thresholds, including $\tau = 0.3$, retained zero training samples.

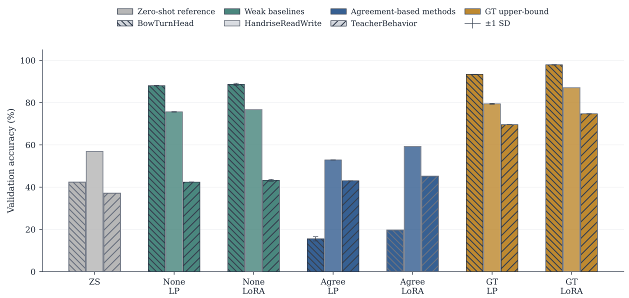

FIGURE 4. Overview of the full evaluation matrix. The plot includes the three zero-shot baselines and all 18 CLIP fine-tuning conditions formed by three datasets, three label sources, and two fine-tuning methods. Solid bars denote repeated or replicated conditions; hatched bars denote single-run references; error bars show ±1 SD when available. ZS = zero-shot CLIP; LP = linear probing; LoRA = low-rank adaptation; Agree = agreement-filtered pseudo-labels. Here, ZS bars follow the baseline zero-shot setting used in Table 7 under the same bbox-crop validation protocol; LP bars use repeated-seed means, whereas LoRA bars combine the targeted five-seed replication values for BowTurnHead and TeacherBehavior under None and GT with protocol-matched seed-42 references elsewhere. Prompt-ensemble zero-shot values are reported separately in the selective-analysis section.

that are less immediately visible from the table alone: unfiltered pseudo-labels substantially improve over zero-shot on BowTurnHead and HandriseReadWrite, agreement filtering is consistently inferior to unfiltered pseudo-labels except for a marginal TeacherBehavior LP gain of 0.63 pp, and ground-truth supervision remains the upper reference across all datasets.

1) Cross-Model Robustness as a Boundary Check

After the main downstream comparison, Tables 9 and 10 summarise validation-set annotation accuracy for the eleven tested MLLMs under the common bbox-crop and index-only prompting protocol. The purpose of this analysis is not to redefine the main pipeline, but to test whether the category-level reliability pattern persists beyond the original two-model pair, across model generations, architectures, and parameter scales.

Three findings emerge from the expanded comparison. First, no single annotator is uniformly best across datasets: Qwen3.5-35B-A3B performs best on BowTurnHead (89.39%), Qwen2 on HandriseReadWrite (70.59%), and Qwen3.5-27B on TeacherBehavior (52.19%). Second, a clear capability boundary exists on the *guide* category: earlier-generation models (Qwen2, Qwen2.5, LLaVA, Gemma3-27B) all score below 25%, while newer-generation models (Qwen3.5-27B, Gemma4-31B) achieve 39–83%, showing that *guide* is a model-capacity bottleneck rather than a task-level impossibility. Third, the *answer* category remains below 23% across all eleven models (the best is Qwen3.6-35B-A3B at 22.4%), consistent with a task-level single-frame observability constraint that even the newest models cannot overcome.

TABLE 9. Eleven-model cross-model robustness check of validation-set annotation accuracy under a common bbox-crop and index-only prompting protocol. Abbreviations: Q2.5-7B = Qwen2.5-VL-7B-Instruct; Q2.5-32B = Qwen2.5-VL-32B-Instruct; G3-27B = Gemma-3-27B-IT; Q3.5-27B = Qwen3.5-27B; Q3.5-35B = Qwen3.5-35B-A3B (MoE); Q3.6-27B = Qwen3.6-27B; Q3.6-35B = Qwen3.6-35B-A3B (MoE, FP8); G4-26B = Gemma4-26B-A4B-it (MoE); G4-31B = Gemma4-31B-it. Qwen2 = Qwen2-VL-7B-Instruct; LLaVA = LLaVA-1.5-7B. "—" indicates the model was not evaluated on that dataset. Values are bbox-level pseudo-label accuracy (%). Best per dataset in bold.

Dataset	Qwen2	LLaVA	Q2.5-7B	Q2.5-32B	G3-27B	Q3.5-27B	Q3.5-35B	Q3.6-27B	Q3.6-35B	G4-26B	G4-31B
BowTurnHead	82.55	14.76	84.86	63.78	75.19	84.93	**89.39**	—	80.35	86.43	86.35
HandriseReadWrite	**70.59**	53.20	51.74	69.11	57.21	—	62.71	—	62.41	56.97	56.95
TeacherBehavior	41.21	37.39	44.86	36.69	44.65	**52.19**	40.33	48.74	41.03	45.15	43.51

TABLE 10. Eleven-model cross-model validation on four key TeacherBehavior categories illustrating the generational boundary (*guide*) and the task-level constraint (*answer*). Values are validation-set bbox-level accuracy (%). *n* = number of validation crops.

Category	n	Qwen2	LLaVA	Q2.5-7B	Q2.5-32B	G3-27B	Q3.5-27B	Q3.5-35B	Q3.6-27B	Q3.6-35B	G4-26B	G4-31B
guide	449	0.22	24.05	0.00	0.89	0.22	52.34	24.50	39.20	12.92	36.04	**82.85**
answer	853	0.00	0.00	1.17	5.16	1.41	17.70	18.87	8.56	**22.39**	0.35	9.38
blackboard-writing	277	98.56	36.82	95.67	—	—	98.19	98.56	**98.92**	97.11	94.58	95.67
screen	1959	69.37	13.78	56.41	—	—	**88.82**	73.51	74.27	68.86	52.78	48.09

Table 10 reports per-category accuracy for four key TeacherBehavior categories that illustrate the generational and task-level boundaries. The *guide* category shows a clear generational divide: earlier-generation models (Qwen2, Qwen2.5, LLaVA, Gemma3-27B) all score below 1% except LLaVA (24.05%, likely a model-specific bias), while newer models reach 12.9–82.9%, with Gemma4-31B achieving 82.9%. *Answer* remains a near-universal failure: all eleven models score below 23%, with the best (Qwen3.6-35B-A3B at 22.4%) still far below usable annotation quality—consistent with a task-level single-frame observability constraint. *Blackboard-writing* and *screen* are reliably annotated across all models (89.9–98.9% and 48.1–88.8%, respectively), consistent with their strong visual signatures. Qwen3.5-27B is the most balanced annotator on TeacherBehavior, with the highest overall accuracy (52.19%) and the best or near-best per-category scores across all four classes shown.

2) Quality-Threshold Experiment with Qwen3.5-27B

The cross-model validation identifies Qwen3.5-27B as the most balanced annotator on TeacherBehavior (val-set accuracy 52.19%, Table 9), substantially outperforming the original Qwen2-VL-7B annotator (41.21%). To test whether this 11pp improvement in annotation quality translates into downstream fine-tuning gains, a targeted experiment was conducted using Qwen3.5-27B pseudo-labels for CLIP fine-

TABLE 11. TeacherBehavior fine-tuning results comparing Qwen2-VL-7B pseudo-labels (annotation accuracy 41.2%) and Qwen3.5-27B pseudo-labels (annotation accuracy 50.3%). The zero-shot CAPE prompt-ensemble baseline is 53.43%. GT upper bounds: LP 69.56%, LoRA 74.71%.

Training condition	Qwen2 (41.2%)	Qwen3.5-27B (50.3%)	Δ
None + LP	42.27	55.29	+13.02
None + LoRA	44.34	53.79	+9.45
Selective + LP	58.37	62.10	+3.73
Selective + LoRA	—	53.20	—
Zero-shot CAPE	*53.43*	*53.43*	—
GT + LP	*69.50*	—	—

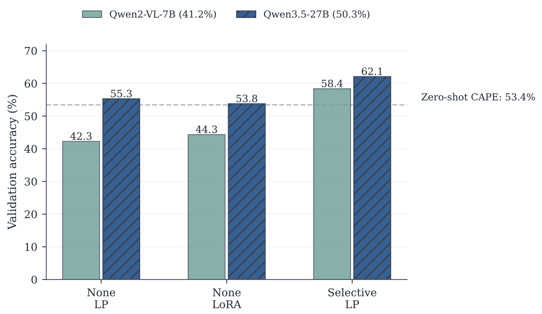

FIGURE 5. TeacherBehavior validation accuracy under Qwen2-VL-7B pseudo-labels (annotation accuracy 41.2%) versus Qwen3.5-27B pseudo-labels (50.3%). The dashed line marks the zero-shot CAPE prompt-ensemble baseline (53.43%). Qwen3.5-27B labels cross the zero-shot threshold for None+LP and Selective+LP, while Qwen2-VL-7B labels remain below it for all conditions.

tuning on TeacherBehavior under the same bbox-crop protocol, training settings, and seed configuration as the main pipeline. The Qwen3.5-27B training-set annotation accuracy is 50.3% (the training labels are used for fine-tuning; validation labels are only used for quality measurement).

Figure 5 visualises the Qwen2 vs. Qwen3.5-27B comparison for the three training conditions where both annotators have results.

Three findings from this experiment are directly relevant to the paper's central question. First, the annotation quality threshold at which pseudo-label fine-tuning begins to exceed the zero-shot baseline lies between 41% and 50%. Qwen2-VL-7B labels (41.2%) produce downstream accuracy below the zero-shot baseline (None+LP: 42.27% vs. 53.43% zero-shot CAPE), while Qwen3.5-27B labels (50.3%) produce accuracy above it (None+LP: 55.29% vs. 53.43% zero-shot CAPE). This pinpoints a practical quality requirement for MLLM-assisted annotation pipelines on TeacherBehavior-level task complexity.

Second, the selective-routing gain scales with annotator quality. Under Qwen2-VL-7B pseudo-labels, selective+LP improves over None+LP by 16.1pp (58.37% vs. 42.27%). Under Qwen3.5-27B pseudo-labels, selective+LP reaches 62.10%, an improvement of 8.67pp over the zero-shot baseline compared with 4.94pp for Qwen2-VL-7B. This confirms that higher-quality base annotations amplify the benefit of anchor-aware supervision routing, even with the same oracle router (Scheme B).

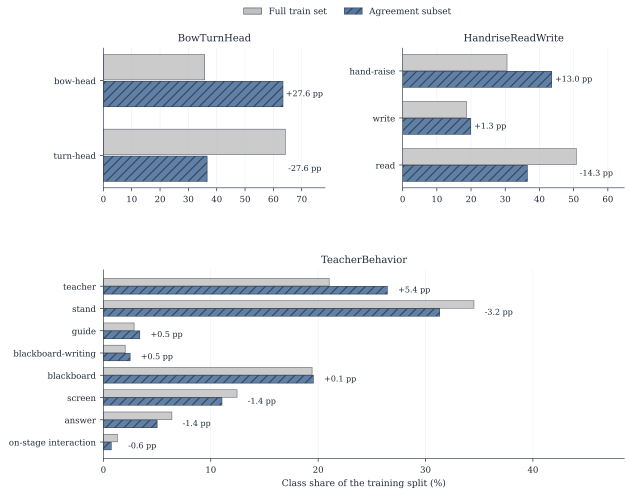

FIGURE 6. Class-composition shift between the full training split and the agreement subset for each dataset. Labels on the agreement bars report percentage-point change relative to the full training distribution, making the direction and magnitude of selection bias explicit.

Third, LoRA remains unstable in the TeacherBehavior selective setting. Selective+LoRA with Qwen3.5-27B labels achieves only 53.20%—below the zero-shot baseline and substantially below selective+LP (62.10%). This reinforces the earlier finding that LoRA is more vulnerable to limited-data overfitting than linear probing when the training subset is small (selective routing retains only high-anchor-class crops).

V. DISCUSSION

A. FINDING 1: AGREEMENT FILTERING IS ASSOCIATED WITH DATA-VOLUME LOSS, DISTRIBUTIONAL BIAS, AND QUALITY CEILING

The central lesson of Table 4, Table 7, and Fig 7 is not simply that agreement filtering can underperform, but that the diagnostics point to three interacting risks. BowTurnHead is consistent with data-volume stress: the filter discards enough examples to push training below a viable retention threshold. HandriseReadWrite shows a different pattern: at moderate retention, random subsampling remains close to the full-data baseline while the agreement subset falls well below it, indicating that biased selection may contribute beyond data reduction alone. TeacherBehavior suggests a third regime: once pseudo-label noise becomes the dominant bottleneck, increasing retention changes little, so filtering does not meaningfully raise the attainable ceiling.

Fig 6 shows that the agreement subset does not preserve the original training distribution. It over-selects *bow-head* in BowTurnHead (35.76% to 63.40%), shifts HandriseReadWrite toward *hand-raise* at the expense of *read* (30.52% to 43.57%; 50.80% to 36.50%), and in TeacherBehavior over-selects *teacher* while under-selecting *stand* (21.03% to 26.45%; 34.51% to 31.32%). The filter changes not only data volume but also the class mix seen during adaptation.

These three regimes support a stronger practical warning than “agreement is imperfect.” Agreement is not a portable denoising rule in this setting because it may simultaneously alter data volume, class composition, and sample difficulty. The five-seed LP averages remain the primary downstream evidence, whereas the retention curves serve as mechanism checks. Taken together, they imply that agreement-filtered subsets should be treated as audited special cases, not as default pseudo-label training data. The cross-model comparison points in the same direction: agreement quality varies materially with the model pair, so simple annotator substitution does not turn the heuristic into a stable strategy.

B. FINDING 2: VISUAL ANCHORING HELPS DIAGNOSE WHEN AGREEMENT FILTERING FAILS

The most useful interpretation of Table 6 is as a diagnostic explanation for the failure patterns above. Annotation reliability depends less on nominal model scale than on whether a class is visually observable in a single crop. Classes such as *blackboard-writing* and *teacher* expose distinctive visual cues that MLLMs can align with pretraining knowledge, whereas categories such as *answer*, *guide*, and *on-stage interaction* depend more on pedagogical intent and discourse context than on a unique frame-level signature. The protocol-matched zero-shot CLIP results are used here as an operational anchoring proxy, not as an independent causal explanation: $\mathrm{AnchorProxy}(y) = \mathrm{Acc}_y^{\mathrm{ZS}}$ helps distinguish categories that are visually legible under the current protocol from those that are not.

The eleven-model boundary check strengthens this interpretation. Although the best annotator changes across datasets, the same intent-dependent failure classes remain near-collapse under model substitution. That consistency suggests a task-level bottleneck, not a defect specific to one annotator pair. A complementary CMC check gives the same pattern in a CLIP-independent form. Class-level CMC is strongly aligned with the zero-shot AnchorProxy on HandriseReadWrite (Pearson $r = 0.996$; Spearman $\rho = 0.500$) and moderately aligned on TeacherBehavior (Pearson $r = 0.573$; Spearman $\rho = 0.439$). BowTurnHead has only two classes and is reported descriptively, not as a correlation analysis. These values do not make the proxy causal, but they reduce the risk that the anchoring interpretation is merely an artefact of a single CLIP zero-shot score. In practical terms, the limiting issue is single-frame observability: if the label cannot be recovered from an isolated crop, changing the annotator is unlikely to solve the problem. Under this reading, visual anchoring is best treated as a screening diagnostic for when pseudo-label supervision is plausible. High-anchor categories are plausible candidates for pseudo-label supervision, whereas low-anchor categories should default to stronger prompting, added audit, human annotation, or richer temporal and multimodal evidence.

More broadly, these findings challenge a common assumption in pseudo-label pipelines: that inter-model agreement is a reliable proxy for label quality. In this setting, agreement

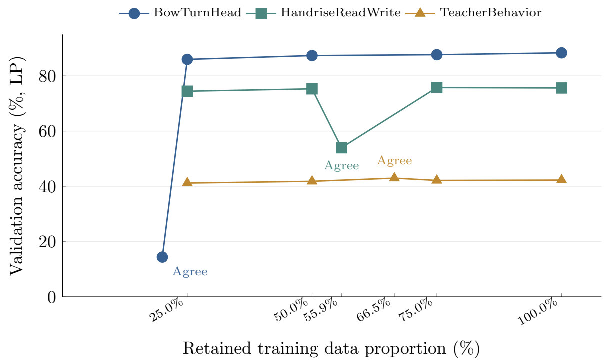

FIGURE 7. Retention-ratio curves for BowTurnHead, HandriseReadWrite, and TeacherBehavior under linear probing with unfiltered pseudo-labels (None strategy), augmented with the agreement-filtered endpoint (labelled “Agree”). All plotted points are fixed-seed diagnostic runs (seed 42), whereas Table 7 reports repeated-seed LP means for the main downstream comparison. The x-axis reports *training-set* retention rates, whereas Table 4 reports *validation-set* retention rates for annotation quality. Random subsampling remains stable above the critical retention threshold for all three datasets. The agreement-selected endpoint falls far below the random-retention trend for BowTurnHead and HandriseReadWrite, whereas TeacherBehavior shows only a small positive deviation.

preferentially retains samples on which both models express the same perceptual bias, instead of samples that are objectively easier to classify. When two annotators inherit similar pretraining corpora or architectural inductive biases, their agreement can reflect shared blind spots instead of genuine label confidence. Under that view, agreement filtering is better understood not as a denoising operation, but as a selection mechanism whose behaviour depends on the model pair and on the visual discriminability of the target categories.

C. FINDING 3: ANNOTATION QUALITY THRESHOLDS AND THE SELECTIVE-ROUTING UPPER BOUND

Two related findings constrain when pseudo-label supervision is likely to be useful. First, the quality-threshold experiment (Table 11) identifies the annotation accuracy range at which pseudo-label fine-tuning crosses the zero-shot baseline. Qwen2-VL-7B labels at 41.2% annotation accuracy produce downstream accuracy below zero-shot (42.27% vs. 53.43% CAPE), while Qwen3.5-27B labels at 50.3% annotation accuracy surpass the baseline (55.29% vs. 53.43%). The critical threshold for TeacherBehavior-level task complexity therefore lies between 41% and 50%, providing a practical quality target for MLLM-assisted annotation pipelines.

Second, the selective-routing diagnostic (Table 12) shows that its benefit scales with base annotation quality. Selective+LP improves over zero-shot by 4.94pp with Qwen2 labels and by 8.67pp with Qwen3.5-27B labels (62.10% vs. 53.43%). This supports the interpretation that anchor-aware supervision routing amplifies, rather than replaces, the benefit of higher-quality annotations. The result remains a boundary diagnostic, not a deployable method, because Scheme B uses oracle routing and the AnchorProxy is coupled to the zero-shot branch.

D. FINDING 4: TASK COMPLEXITY MODERATES THE BENEFIT OF PSEUDO-LABEL FINE-TUNING

The downstream comparison answers the central question conditionally. Unfiltered pseudo-labels can beat zero-shot CLIP when the task is visually grounded enough that label noise does not erase the training signal. That is the pattern on BowTurnHead and, more moderately, on HandriseReadWrite. The same strategy is far less persuasive on TeacherBehavior, where the label semantics depend more heavily on intent and interaction context.

Ground-truth fine-tuning clarifies the distinction. Clean labels remain strongly useful across all three datasets, so the weak pseudo-label result on TeacherBehavior should not be read as evidence that the task itself is resistant to adaptation. It is instead evidence that task complexity and label semantics jointly determine whether pseudo-label supervision is worth using. The auxiliary baselines sharpen this point. Confidence filtering recovers the unfiltered LP regime on BowTurnHead and HandriseReadWrite but fails to retain any TeacherBehavior samples under the evaluated thresholds, consistent with uniformly weak CLIP confidence on that taxonomy. Conversely, the train-split teacher–student baseline raises TeacherBehavior to 65.06%, close to the 69.56% GT LP reference, while falling to 68.08% on HandriseReadWrite. The useful conclusion is not that self-training is universally better, but that TeacherBehavior's weak direct pseudo-label result is primarily a label quality problem under this protocol. A practical decision rule follows: pseudo-label fine-tuning is most plausible when classes are visually grounded and semantically stable; as intent dependence increases, selective human annotation becomes more valuable.

E. FINDING 5: THE ATTAINABLE PERFORMANCE CEILING IS MORE STRONGLY LOCKED BY LABEL QUALITY THAN BY OPTIMISATION PATH

The five-seed LoRA replications sharpen the same conclusion from a different angle. What changes most across conditions is not only mean accuracy but the ceiling itself. Ground-truth supervision produces both higher validation scores and much lower cross-seed variance, whereas pseudo-label supervision keeps the model in a noisier regime. Across the replicated BowTurnHead and TeacherBehavior settings, the mean cross-seed SD is 0.107% under GT supervision versus 0.495% under pseudo-label supervision. On TeacherBehavior in particular, the pseudo-label setting can fit the training set almost perfectly while validation remains trapped in a narrow band. That pattern is more consistent with memorising inconsistent decision boundaries than with simply choosing the wrong checkpoint or optimisation schedule.

This matters because it limits one easy counterargument to the main results. The weak pseudo-label outcome cannot be explained away by saying that LoRA merely needed more careful tuning. Under this single-frame protocol, optimisation changes the training trajectory more than it changes the attainable validation regime. The remaining single-run LoRA cells should still be read cautiously, but the replicated seeds are sufficient to support a narrower conclusion: label quality constrains both performance and stability more strongly than optimisation path in the settings where the paper makes its strongest claims.

TABLE 12. Comparison of annotation strategies on TeacherBehavior (validation accuracy, %). The selective strategy uses pseudo-labels for high-anchor classes and zero-shot CLIP for low-anchor classes. High-anchor: blackboard-writing, teacher, screen, blackboard. Low-anchor: guide, answer, on-stage interaction, stand. The zero-shot baseline in this table uses the prompt-ensemble setting. Selective routing uses ground-truth category as oracle routing variable at evaluation time and represents a diagnostic upper bound, not a deployable pipeline or a deployable-method comparison.

Strategy	Overall (8-cls)	High-anchor (4-cls)	Low-anchor (4-cls)
Zero-shot CLIP (prompt ensemble)	53.43	—	—
None + LP (full pseudo)	42.38	—	—
Oracle selective + LP	58.37	72.74	38.44
GT + LP (upper bound)	69.56	—	—

F. TEACHERBEHAVIOR CASE STUDY: WHAT THE SELECTIVE-ROUTING GAIN DOES AND DOES NOT SHOW

The TeacherBehavior case study makes the boundary condition concrete. When agreement-based selection is unreliable, category-aware routing can still be informative as an upper-bound diagnostic, even though the present Scheme B implementation is not directly deployable. Table 12 summarises the result.

The gain comes from the high-anchor portion of the taxonomy, while the low-anchor classes remain better handled by the zero-shot branch. That is precisely why the result should not be interpreted as a new universal pipeline. It depends on oracle routing and on a fine-tuning setup that preserves the zero-shot branch.

This constraint also explains why LoRA is a poor fit for Scheme B. Once the shared encoder is altered, the fallback branch for low-anchor categories is no longer stable. The case study is most useful as a design implication. If a taxonomy mixes visually explicit and intent-dependent classes, selective routing may be worth exploring, but only with an explicit routing mechanism and without presenting the oracle result as deployment-ready.

G. LIMITATIONS

Several limitations should be noted. First, the main annotation pipeline evaluates two 7B-scale locally deployable MLLMs, and the expanded cross-model robustness analysis spans eleven models across seven families: Qwen2-VL-7B, LLaVA-1.5-7B, Qwen2.5-VL-7B-Instruct, Qwen2.5-VL-32B-Instruct, Qwen3.5-27B, Qwen3.5-35B-A3B, Qwen3.6-27B, Qwen3.6-35B-A3B, Gemma-3-27B-IT, Gemma4-26B-A4B-it, and Gemma4-31B-it. This broader evidence reduces concern that the reported failure patterns are unique to one particular model pair, but broader validation across more datasets, prompt regimes, and multimodal model families remains necessary. Whether larger-scale or video-aware models materially improve annotation reliability for intent-dependent categories remains an open question. Second, the annotation pipeline uses image-level prompts without chain-of-thought reasoning or few-shot examples; more sophisticated prompting strategies may improve accuracy on intent-dependent categories. In particular, the index-only output constraint used here standardises parsing but does not expose uncertainty, so prompt sensitivity and abstention-aware prompting remain open questions. Third, the study uses a single benchmark (SCB) from Chinese K–12 classrooms; generalisability to other educational contexts and cultural settings requires further investigation. Fourth, the selective annotation experiment evaluates only linear probing; LoRA is incompatible with Scheme B due to encoder-level interference with the zero-shot classifier, and alternative architectures that decouple the fine-tuned head from the shared encoder (e.g., adapter layers, separate encoder branches) are left for future work. Fifth, the main LP conditions are averaged over five random seeds (all datasets: five seeds, 43–47), but the LoRA replication now covers BowTurnHead and TeacherBehavior under None and GT across five seeds (43–47). However, the remaining LoRA settings and several diagnostic analyses remain single-run protocol-matched evaluations. The reported differences should still be interpreted as controlled diagnostic comparisons rather than as full estimates of training variance across all conditions. The study does not support formal significance claims for small differences in conditions that remain single-run, and future work should extend repeated-seed evaluation to the remaining LoRA and auxiliary diagnostic analyses, together with bootstrap- or permutation-based uncertainty estimates. The eleven-model robustness analysis and the Qwen3.5-27B quality-threshold experiment partly strengthen external validity with respect to annotator choice, but they do not remove the internal validity limitations in the downstream fine-tuning matrix. The fine-tuning hyperparameters are also held fixed across the main comparisons, so systematic learning-rate sensitivity is not evaluated in this study. Sixth, the paper treats visual anchoring as a diagnostic concept that is operationalised here with a task-specific proxy ($\mathrm{AnchorProxy}(y) = \mathrm{Acc}_y^{\mathrm{ZS}}$), rather than a universal quantitative variable. Because this proxy is itself derived from classifier performance, it should be interpreted as an operational diagnostic, not a standalone explanatory construct. The CMC analysis adds a CLIP-independent consistency check, but it is still descriptive and does not establish a causal measurement of anchoring. Future work should evaluate more general category-level indicators and test whether their association with pseudo-label accuracy is stable across datasets and model families. Seventh, the practical training-time baselines included here remain deliberately narrow. Confidence filtering uses a CLIP-assisted probability proxy because the index-only MLLM annotation protocol does not expose calibrated model uncertainty, and the teacher–student baseline uses one linear-probe self-training design with a stratified 10% labeled subset of the training split. Prompt-adaptation methods such as CoOp/CoCoOp, calibrated MLLM abstention, and more elaborate semi-supervised objectives remain outside this comparison. Eighth, although the retention-ratio analysis and distribution-shift metrics (total-variation distance, KL divergence) provide partial controls for data-volume and sampling-bias effects, the three failure mechanisms identified for agreement filtering—data-volume collapse, distributional bias, and quality ceiling—are not fully causally isolated in the current experimental design. Agreement-filtered subsets differ from random subsets simultaneously in data volume, class distribution, sample difficulty, and visual diversity, so the present mechanism attribution should be read as diagnostically motivated, not causally verified. Controlled experiments that independently vary these factors—for example, class-balanced resampling matched to the agreement subset distribution, or difficulty-stratified retention curves—are left for future work.

H. REPRODUCIBILITY STATEMENT

The repository implements these analysis modules using the following scripts at the repository root: zero-shot evaluation (baseline and prompt-ensemble settings, `step0b_cape_zeroshot.py`), MLLM annotation (`step1_llm_annotate.py` and `cross_model_annotate.py`), filtering analysis (`step2_filter_analysis.py`), CLIP fine-tuning (`step3_clip_finetune.py`), selective annotation (`step4_selective_annotation.py`), retention-ratio curve (`step5_retention_curve.py`), and repeated-seed aggregation (`summarize_repeated_seeds.py`). The auxiliary strategy and consistency analyses are implemented in `confidence_filtering.py`, `teacher_student_self_training.py`, and `cross_model_consistency.py`. The submitted artifact includes `environment.yml` and `requirements.txt` for environment reconstruction. The experiment snapshot used for manuscript preparation is identified by repository commit `e21ae3d09c8d1577eeb75d1a29e2661cf757e8`
the archived artifact should preserve the exact submitted snapshot together with the generated result files. LP main-matrix results are averaged over repeated seeds (all datasets: five seeds, 43–47); targeted LoRA replications for BowTurnHead and TeacherBehavior under None and GT use seeds 43–47, while remaining LoRA and selected diagnostics are single-seed/fixed-seed evaluations (seed 42). Table-level provenance is as follows: annotation-quality and confusion tables are generated by the filtering-analysis module, the main fine-tuning matrix by the CLIP fine-tuning and repeated-seed summary modules, the selective-routing table by the selective-annotation module, the auxiliary-baseline table by the confidence-filtering and teacher–student modules, and the retention-ratio figure by the retention-curve module. The SCB dataset is publicly available [33]. The code repository and run configurations used for the analysis are publicly available at https://github.com/zhanglizhuo/llm-annotation (accessed on 27 April 2026), including the pipeline scripts, archived shell entry points, and result directories referenced in this paper. For reproducibility review, the repository should remain publicly accessible and preserve per-module run recipes for the main tables and figures.

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VI. CONCLUSIONS

The paper asked a practical question: when can MLLM-generated pseudo-labels improve classroom behavior recognition, and when does agreement filtering make them worse instead of better? Within the SCB benchmark, the answer is conditional rather than global. Pseudo-labels help when the target classes are visually anchored enough that single-frame crops preserve the evidence needed for labeling. They do not help reliably when the taxonomy is dominated by intent-dependent classes, and cross-model agreement is not a safe remedy because it may remove training volume, distort class composition, and preserve shared bias instead of correctness.

The evidence supporting that answer is consistent across the manuscript. Category-level annotation results show a sharp divide between visually anchored classes and intent-dependent classes. The repeated-seed LP matrix shows that unfiltered pseudo-labels can outperform zero-shot CLIP on visually grounded datasets, whereas the clearest counterexample appears on BowTurnHead, where Agreement+LP falls to 15.53% $\pm$ 1.00% while None+LP reaches 88.02% $\pm$ 0.12%. The targeted LoRA replications indicate that label quality constrains not only attainable accuracy but also run-to-run stability. The quality-threshold experiment locates the critical annotation accuracy range at which pseudo-label fine-tuning crosses the zero-shot baseline on TeacherBehavior (between 41% and 50%), and the expanded eleven-model cross-model validation reveals a generational boundary on *guide* while confirming that *answer* remains below 23% across all tested models. The CMC diagnostic, the auxiliary confidence-filtering and teacher–student baselines, the oracle Scheme B diagnostic, and the eleven-model robustness check then bound the claim: pseudo-label supervision is most plausible for high-anchor categories, not as a universal replacement for human annotation or temporal and multimodal evidence.

The practical implication is not “filter harder” but “screen the task first.” For SCB-like single-frame classroom sensing, a conservative workflow is to audit category anchoring before committing annotation budget, avoid agreement filtering as a default denoising rule, select an annotator with task-level accuracy above the zero-shot threshold (approximately 50% for TeacherBehavior-level complexity), use pseudo-label supervision mainly on high-anchor categories, and reserve low-anchor categories for stronger zero-shot prompts, temporal or multimodal sensing, or human labelling. Under this interpretation, the contribution of the paper is a reproducible reliability-audit protocol for deciding when MLLM pseudo-labeling is useful in classroom behavior recognition, together with diagnostic evidence on the main failure modes that make it unreliable in this setting.

AUTHOR CONTRIBUTIONS

Conceptualization, Y.M. and L.Z.; methodology, Y.M. and L.Z.; software, L.Z.; validation, Y.M. and L.Z.; formal analysis, Y.M.; investigation, Y.M.; data curation, Y.M. and L.Z.; visualization, L.Z.; writing—original draft preparation, Y.M.; writing—review and editing, Y.M. and L.Z.; supervision, L.Z.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

DATA AND CODE AVAILABILITY

The three SCB sub-datasets used in this study, plus the excluded single-class Discuss set, are publicly available on Hugging Face at https://huggingface.co/datasets/wintonYF/SCB-Dataset (accessed on 27 April 2026). These datasets are externally released resources and are not owned by this study. The experiment code and archived run configurations are available at https://github.com/zhanglizhuo/llm-annotation (accessed on 27 April 2026). The repository was publicly accessible at the time of access.

ACKNOWLEDGMENT

The authors thank wintonYF and all contributors who created and maintained the public SCB dataset releases used in this study.

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